

Q1 - 25 July - Shift 2

A ball is projected from the ground with a speed 15 ms^{-1} at an angle θ with horizontal so that its range and maximum height are equal, then ' $\tan \theta$ ' will be equal to

- (A) $\frac{1}{4}$ (B) $\frac{1}{2}$
(C) 2 (D) 4

Space for your notes:

Q2 - 26 July - Shift 1

Two projectile thrown at 30° and 45° with the horizontal respectively, reach the maximum height in same time. The ratio of their initial velocities is

- (A) $1:\sqrt{2}$ (B) $2:1$
(C) $\sqrt{2}:1$ (D) $1:2$

Space for your notes:

Q3 - 26 July - Shift 1

If the initial velocity in horizontal direction of a projectile is unit vector $\hat{i}$ and the equation of trajectory is $y = 5x(1-x)$. The y component vector of the initial velocity is _____ $\hat{j}$

(Take $g = 10 \text{ m/s}^2$)

Space for your notes:

Q4 - 27 July - Shift 1

Questions

MathonGo

A ball of mass m is thrown vertically upward.

Space for your notes:

Another ball of mass $2m$ is thrown at an angle θ with the vertical. Both the balls stay in air for the same period of time. The ratio of the heights

attained by the two balls respectively is $\frac{1}{x}$. The

value of x is _____.

Q5 - 27 July - Shift 2

A body of mass 10 kg is projected at an angle of 45° with the horizontal. The trajectory of the body is observed to pass through a point $(20, 10)$. If T is the time of flight, then its momentum vector, at time $t = \frac{T}{\sqrt{2}}$, is _____

Space for your notes:

[Take $g = 10\text{ m/s}^2$]

- (A) $100\hat{i} + (100\sqrt{2} - 200)\hat{j}$
- (B) $100\sqrt{2}\hat{i} + (100 - 200\sqrt{2})\hat{j}$
- (C) $100\hat{i} + (100 - 200\sqrt{2})\hat{j}$
- (D) $100\sqrt{2}\hat{i} + (100\sqrt{2} - 200)\hat{j}$

Q6 - 28 July - Shift 1

#MathBoleTohMathonGo

Questions

MathonGo

A NCC parade is going at a uniform speed of 9 km/h under a mango tree on which a monkey is sitting at a height of 19.6 m. At any particular instant, the monkey drops a mango. A cadet will receive the mango whose distance from the tree at time of drop is :

(Given $g = 9.8 \text{ m/s}^2$)

- (A) 5 m (B) 10 m
(C) 19.8 m (D) 24.5 m

Space for your notes:

Q7 - 28 July - Shift 2

At time $t = 0$ a particle starts travelling from a height $7\hat{z}$ cm in a plane keeping z coordinate constant. At any instant of time it's position along the x and y directions are defined as $3t$ and $5t^3$ respectively. At $t = 1$ s acceleration of the particle will be

- (A) $-30\hat{y}$ (B) $30\hat{y}$
(C) $3\hat{x} + 15\hat{y}$ (D) $3\hat{x} + 15\hat{y} + 7\hat{z}$

Space for your notes:

Q8 - 29 July - Shift 1

#MathBoleTohMathonGo

A ball is projected with kinetic energy E , at an angle of 60° to the horizontal. The kinetic energy of this ball at the highest point of its flight will become :

- (A) Zero (B) $\frac{E}{2}$
(C) $\frac{E}{4}$ (D) E

Space for your notes:

Q9 - 29 July - Shift 1

An object is projected in the air with initial velocity u at an angle θ . The projectile motion is such that the horizontal range R , is maximum. Another object is projected in the air with a horizontal range half of the range of first object. The initial velocity remains same in both the case. The value of the angle of projection, at which the second object is projected, will be _____ degree.
(Mark the smallest angle possible)

Space for your notes:

Answer Key

Q1 (D)

Q2 (C)

Q3 (5)

Q4 (1)

Q5 (D)

Q6 (A)

Q7 (B)

Q8 (C)

Q9 (15)

#MathBoleTohMathonGo

Q1 (D)

$$R = H$$

$$\frac{2v_x \times v_y}{g} = \frac{v_y^2}{2g}$$

$$v_x = \frac{v_y}{4}; u \cos \theta = \frac{u \sin \theta}{4}$$

$$\tan \theta = 4$$

Q2 (C)

Time taken to reach maximum height

$$t = \frac{u \sin \theta}{g}$$

$$\therefore \frac{u_1 \sin \theta_1}{g} = \frac{u_2 \sin \theta_2}{g}$$

$$\Rightarrow u_1 \sin 30 = u_2 \sin 45$$

$$\Rightarrow \frac{u_1}{u_2} = \frac{1/\sqrt{2}}{1/2} = \frac{\sqrt{2}}{1}$$

Q3 (5)

$$u_x = 1$$

$$y = 5x(1-x)$$

$$\frac{dy}{dt} = 5 \frac{dx}{dt} - 10x \frac{dx}{dt}$$

For initial y-component of velocity

$$u_y = \left(\frac{dy}{dt} \right)_{x=0} \Rightarrow 5(1) = 5$$

$$\vec{u}_y = 5\hat{j}$$

Q4 (I)

Time of flight is same

$\Rightarrow$ vertical component of velocity is same

$\Rightarrow H_{\max}$ is same

Q5 (D)

$$y = x - \frac{10x^2}{2u^2 \left(\frac{1}{2}\right)} \Rightarrow 10 = 20 - \frac{(10)(100)}{u^2}$$

$$u = 20$$

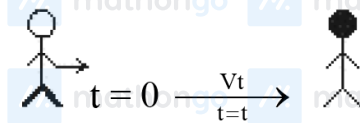
$$T = \frac{(2)(20)}{\sqrt{2}(10)} = 2\sqrt{2}$$

$$\vec{v} = 10\sqrt{2} \hat{i} + (10\sqrt{2} - 10(2)) \hat{j}$$

$$\text{Momentum } \vec{p} = M \vec{v} = 100\sqrt{2} \hat{i} + (100\sqrt{2} - 200) \hat{j}$$

Q6 (A)

Monkey



$$\begin{aligned} \text{Time taken by mango} &= \sqrt{\frac{2n}{g}} \\ &= \sqrt{\frac{2 \times 19.6}{9.8}} = 2 \text{ second} \end{aligned}$$

Distance = vt

$$= 9 \times \frac{5}{18} \times 2 = 5\text{m}$$

Q7 (B)

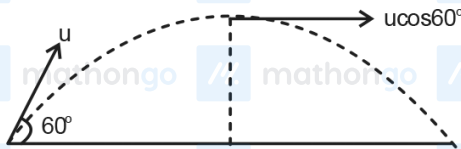
#MathBoleTohMathonGo

$$\vec{r} = 3t\hat{i} + 5t^3\hat{j} + 7t\hat{k}$$

$$\frac{d^2\vec{r}}{dt^2} = 30t\hat{j}$$

$$\text{At } t = 1 \Rightarrow \frac{d^2\vec{r}}{dt^2} = 30\hat{j}$$

Q8 (C)



$$E = \frac{1}{2}mu^2$$

$$\text{At Highest point, Velocity } V = u \cos 60^\circ = \frac{u}{2}$$

$$\therefore \text{K.E at topmost point} = \frac{1}{2}m\left(\frac{u}{2}\right)^2 = \frac{E}{4}$$

Q9 (15)

#MathBoleTohMathonGo

$$R_{\max} = \frac{u^2 \sin 2(45^\circ)}{g} = \frac{u^2}{g}$$

$$\frac{R}{2} = \frac{u^2}{2g} = \frac{u^2 \sin 2\theta}{g}$$

$$\sin 2\theta = \frac{1}{2}$$

$$2\theta = 30^\circ, 150^\circ$$

$$\theta = 15^\circ, 75^\circ$$

Ans. 15, 75